Table 1

Odds ratios for breast cancer and total PCBs in relation to CYP1A1 M1 genotypes in African American and white women

Total PCBs[a]	*CYP1A1* M1 genotype	All participants		Premenopausal		Postmenopausal	
		Patients/controls	OR (95% CI)[b]	Patients/controls	OR (95% CI)[b]	Patients/controls	OR (95% CI)[b]
African Americans							
<0.430	Non-M1	66/75	Referent	46/51	Referent	20/24	Referent
≥0.430	Non-M1	75/67	1.5 (0.9–2.5)	21/16	1.9 (0.9–4.2)	54/51	1.0 (0.5–2.3)
<0.430	Any M1	42/46	1.0 (0.6–1.7)	35/33	1.2 (0.6–2.2)	7/13	0.6 (0.2–1.8)
≥0.430	Any M1	59/54	1.4 (0.8–2.5)	17/14	1.7 (0.8–4.1)	42/40	1.0 (0.5–2.3)
ICR (95% CI)			0.0 (-0.9,0.9)		-0.3 (-2.2,1.6)		0.4 (-0.5,1.3)
Whites							
<0.349	Non-M1	174/133	Referent	118/83	Referent	56/50	Referent
≥0.349	Non-M1	122 148	0.7 (0.5–1.0)	38/43	0.6 (0.4–1.1)	84/105	0.8 (0.5–1.3)
<0.349	Any M1	45/44	0.8 (0.5–1.2)	29/31	0.7 (0.4–1.2)	16/13	1.1 (0.5–2.5)
≥0.349	Any M1	29/32	0.8 (0.4–1.4)	11/10	0.7 (0.3–1.9)	18/22	0.8 (0.4–1.7)
ICR (95% CI)			0.4 (-0.2,0.9)		0.5 (-0.3,1.2)		0.0 (-1.1,1.0)

[a]Lipid-adjusted total imputed PCBs (ng/ml).
[b]Odds ratios adjusted for age and sampling fractions.
Any M1, M1/M1 or M1/non-M1; CI, confidence interval; ICR, interaction contrast ratio; non-M1, non-M1/non-M1.

Table 2

Odds ratios for breast cancer and total PCBs in relation to CYP1A1 M2 genotypes in white women

Total PCBs[a]	*CYP1A1* M2 genotype	All participants		Premenopausal		Postmenopausal	
		Patients/controls	OR (95% CI)[b]	Patients/controls	OR (95% CI)[b]	Patients/controls	OR (95% CI)[b]
<0.349	Non-M2	210/158	Referent	141/98	Referent	69/60	Referent
≥0.349	Non-M2	138/166	0.7 (0.5–1.0)	45/51	0.6 (0.4–1.0)	93/115	0.8 (0.5–1.2)
<0.349	Any M2	11/20	0.4 (0.2–0.8)	8/16	0.3 (0.1–0.8)	3/4	0.7 (0.1–3.1)
≥0.349	Any M2	15/14	0.9 (0.4–1.9)	6/2	2.1 (0.4–10.6)	9/12	0.7 (0.3–1.9)
ICR (95% CI)			0.8 (0.1, 1.6)		2.1 (-1.3,5.5)		0.3 (-0.9,1.5)

[a]Lipid-adjusted total imputed PCBs (ng/ml).
[b]Odds ratios adjusted for age and sampling fractions.
Any M1, M1/M1 or M1/non-M1; CI, confidence interval; ICR, interaction contrast ratio; non-M1, non-M1/non-M1.

exposure greater than or equal to the median among premenopausal African American women regardless of *CYP1A1* M1 genotype. There was no evidence for strong joint effects of PCBs and *CYP1A1* M1-containing genotypes. ICRs were close to zero in African American and white women, and LRTs were not statistically significant. Joint effects for PCBs and *CYP1A1* M2-containing genotypes among white women are presented in Table 2. Greater than additive joint effects were observed among all participants, with an ICR greater than zero and an associated *P* value of 0.03. Although imprecise, joint effects seemed to be stronger among premenopausal than among postmenopausal women. *P* values for LRTs were 0.02 among all participants, 0.007 among premenopausal women, and 0.66 among postmenopausal women.

ORs estimating the joint effects of PCBs and *CYP1A1* M3 genotype among African American women are presented in Table 3. ORs were slightly elevated for women with elevated PCB levels and *CYP1A1* M3-containing genotypes. An ICR greater than zero was observed in postmenopausal women. *P* values for LRTs were 0.10 in all participants, 0.71 among premenopausal women, and 0.11 in postmenopausal women. Joint effects of PCBs and *CYP1A1* M4 genotypes among white women are presented in Table 4. An ICR greater than zero ($P = 0.03$) was observed among